

ACADEMIC DETAILS

Degree	Specialization	Institute	Year	CPI
B.Tech.	Mechanical Engineering	IIT Gandhinagar	2023-Present	8.43
Class XII	Physics, Chemistry, Maths	Rajiv Gandhi Junior College	2020-2022	84.33
Class X		R.J Thakur English Medium School	2010-2020	94.8

PROJECTS

- **Image Classification and Representation Learning using MLP's and CNNs**

[Project Link](#)

[Aug'25-Nov'25]

- Developed an MLP-based image classifier for MNIST and benchmarked it against Random Forest and Logistic Regression using accuracy, F1-score, and confusion-matrix analysis.
 - Compared a custom CNN with pretrained MobileNetV2 and EfficientNet-B0, analyzing classification performance, model size, computational efficiency, and inference-time tradeoffs across architectures.
 - Visualized learned representations using t-SNE before and after training and evaluated cross-domain generalization on Fashion-MNIST to analyze the learned feature space.
 - Tools/Concepts: PyTorch, CNNs, MLPs, Pretrained CNNs, t-SNE, Representation Learning, Model Evaluation
- **SVD-Based Image Watermarking and Robustness Analysis**

[Project Link](#)

[Aug'26]

- Implemented SVD using eigendecomposition and applied it independently to RGB channels for QR-code watermark embedding and recovery through controlled singular-value modification across image channels.
 - Analyzed the trade-off between watermark strength and image distortion using PSNR, identifying intermediate watermark strengths that balance visual quality and recovery quality effectively across different settings.
 - Evaluated watermark robustness under Gaussian noise and JPEG compression using PSNR and QR decoding, analyzing degradation across varying noise levels and compression quality settings systematically.
 - Tools/Concepts: Python, NumPy, OpenCV, SVD, Eigenvalue Decomposition, Image Processing, PSNR, Noise Robustness
- **Template and Feature Based Object Detection**

[Project Link](#)

[Aug'26]

- Implemented image processing primitives including convolution, Gaussian and median filtering, Sobel edge detection, Harris corners, and structural similarity metrics using NumPy from scratch.
 - Built normalized cross-correlation template matching with integral images for efficient search, extending detection across image scales and evaluating robustness to noise, grayscale conversion, and occlusion.
 - Developed a feature-based detector using scale-space keypoints, dominant orientation, local descriptors, descriptor matching, and consensus-based geometric localization for object detection under scale and rotation changes.
 - Tools/Concepts: Python, OpenCV, Image Filtering, Edge Detection, Harris Corners, Feature Descriptors, Feature Matching
- **Human Activity Recognition using Decision Trees**

[Project Link](#)

[Aug'25-Nov'25]

- Implemented a Decision Tree classifier from scratch using entropy, Gini impurity, and MSE-based splitting, supporting both discrete classification and continuous output prediction with recursive tree construction.
 - Processed accelerometer data using raw and TSFEL-engineered time-series features, applying PCA for dimensionality reduction and comparing feature representations across different model configurations.
 - Evaluated classification performance across tree depths using nested cross-validation, precision, recall, and model complexity, analyzing the trade-off between predictive performance and interpretability.
 - Tools/Concepts: Scikit-learn, PCA, TSFEL, Decision Trees, Feature Engineering, Time-Series Analysis, Cross-Validation
- **Multimodal Sentiment Analysis and smart data labelling**

[Project Link](#)

[Jan'26-April'26]

- Built a cost-efficient sentiment analysis pipeline combining human annotation, weak supervision, active learning, and LLM-based labeling, implementing Fleiss' Kappa, heuristic labeling, and uncertainty based sample selection.
 - Implemented Fleiss' Kappa, heuristic labeling functions with Snorkel, and least-confidence and entropy-based active learning to quantify label quality, reduce annotation effort, and identify uncertain samples.
 - Developed text and speech augmentation pipelines using synonym replacement, back-translation, LLM-generated reviews, and synthetic speech; evaluated generated data using similarity, sentiment consistency, and speech metrics.
 - Tools/Concepts: PyTorch, Snorkel, Label Studio, Active Learning, Weak Supervision, LLMs, Data Augmentation, Whisper

ACHIEVEMENTS

- Won the 3rd Prize at Invention Factory, IIT Bombay – a six-week summer program where students identify problems, develop solutions, build prototypes and pitch their ideas to industry professionals. Successfully filed an **Indian patent**, Provisional Patent application number: 202411073551 and **US provisional Patent** (USPTO Application: 63/811,945).

• Received **Academic Citation at IIT Gandhinagar** for outstanding academic performance in 2nd year, Semester I of the academic year 2024-25.

POSITIONS OF RESPONSIBILITY

- **Secretary, Mean Mechanics (Robotics Club), IIT Gandhinagar**

[Jun'25 - April'26]
- **General Member, Technical Council, IIT Gandhinagar**

[Sep'24 - Apr'25]
- **Sponsorship Executive, TEDx IIT Gandhinagar**

[May'24 - March'25]
- **General Member, Career Development Services (CDS), IIT Gandhinagar**

[Sep'24 - Feb'25]